



Graphs in Machine Learning

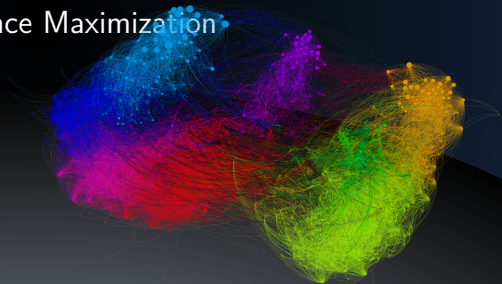
Submodularity: Applications

Product Placement and Influence Maximization

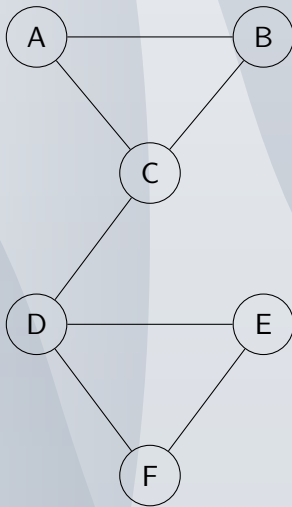
Michal Valko

Isara Labs

Partially based on material by: Andreas Krause,
Branislav Kveton, Michael Kearns

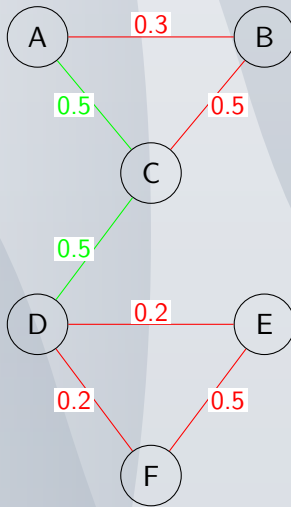


Success story #1 Product placement - solution



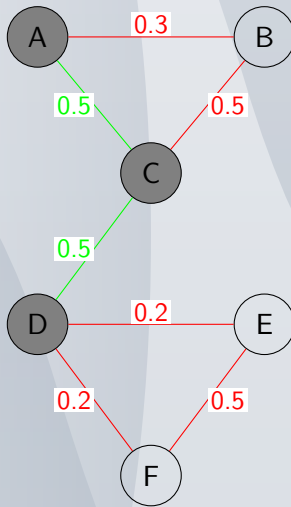
Key idea: Flip coins c in advance $\rightarrow$ “live” edges

Success story #1 Product placement - solution



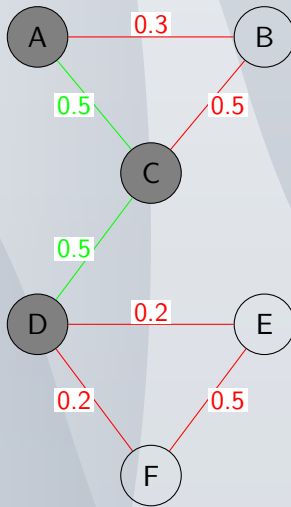
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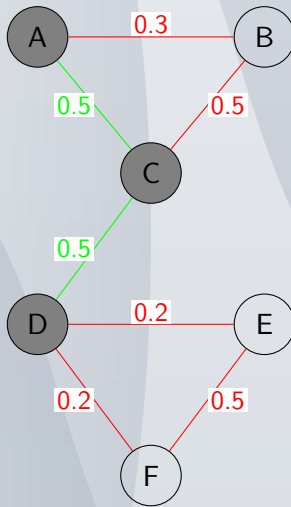
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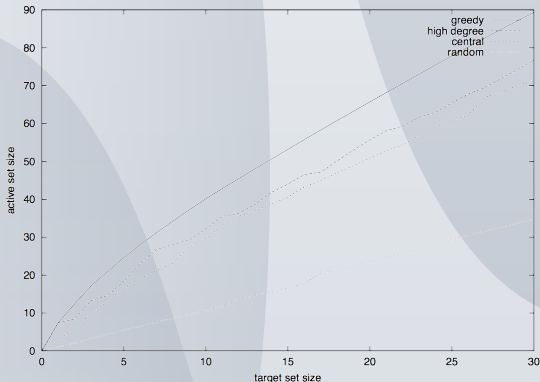
Success story #1 Product placement - solution



Key idea: Flip coins c in advance $\rightarrow$ “live” edges

Success story #1 Product placement - comparison

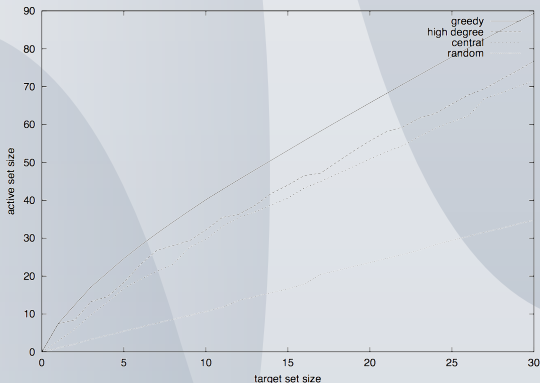
influence on the ArXiv/Physics co-authorship graph



Source: Kleinberg (2000)

Success story #1 Product placement - comparison

influence on the ArXiv/Physics co-authorship graph



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greedy approximation does better than the centrality measures

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<https://misovalko.github.io/mva-ml-graphs.html>